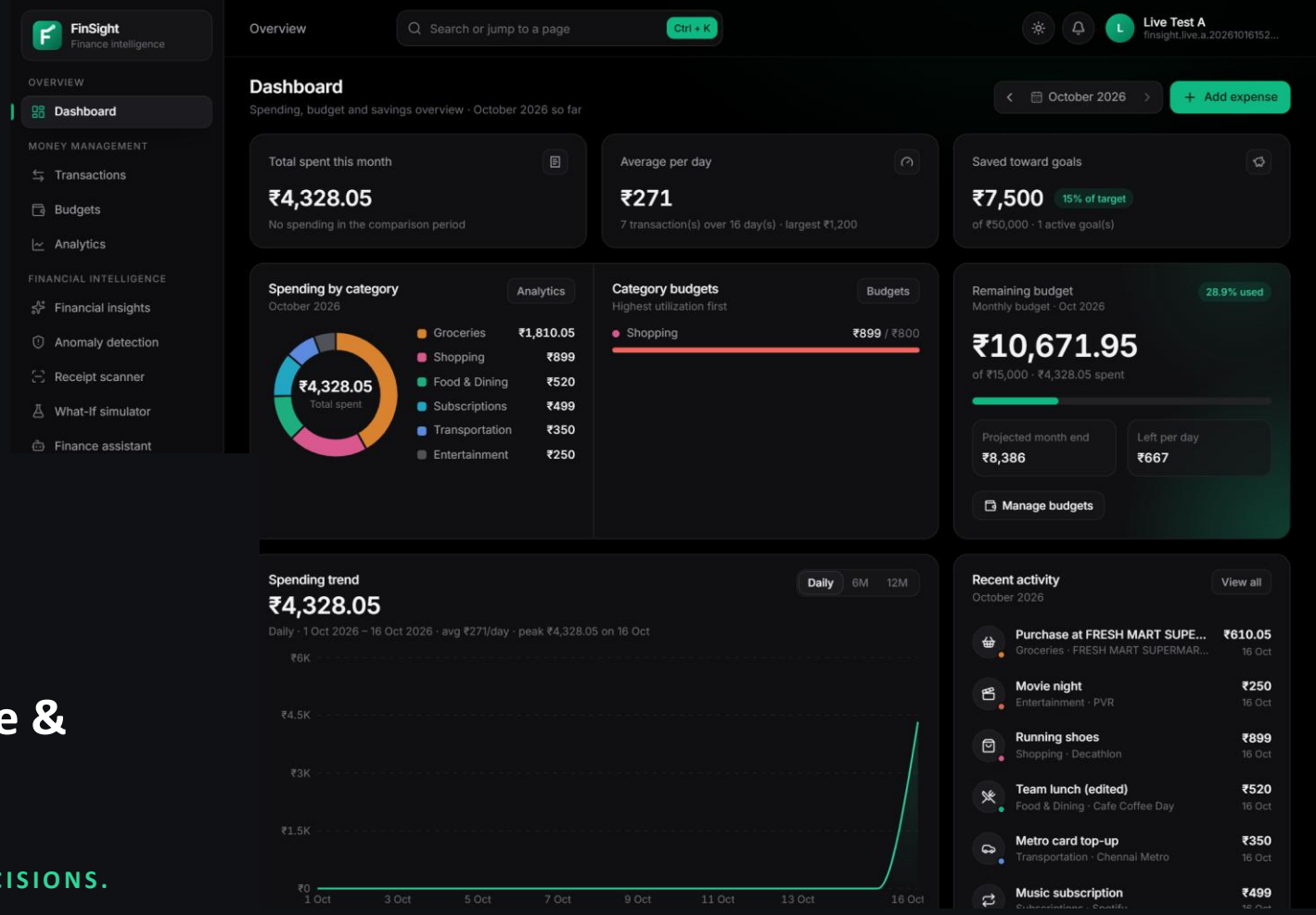
FINSIGHT · AAKAR GUPTA

FINSIGHT

Cloud-Based Personal Finance Intelligence & Expense Analytics Platform

UNDERSTAND YOUR SPENDING. MAKE SMARTER FINANCIAL DECISIONS.



Aakar Gupta
Registration No. 24BRS1321
Vellore Institute of Technology, Chennai

LIVE ON AWS · AP-SOUTH-1 (MUMBAI)
zkht2b4ib3.execute-api.ap-south-1.amazonaws.com
API Gateway · Lambda · DynamoDB · S3 · Cognito

PROBLEM AND MOTIVATION

WHY PEOPLE LOSE TRACK OF THEIR MONEY

Low visibility

Spending is split across UPI, cards and cash, so totals and trends are hidden.

Manual entry

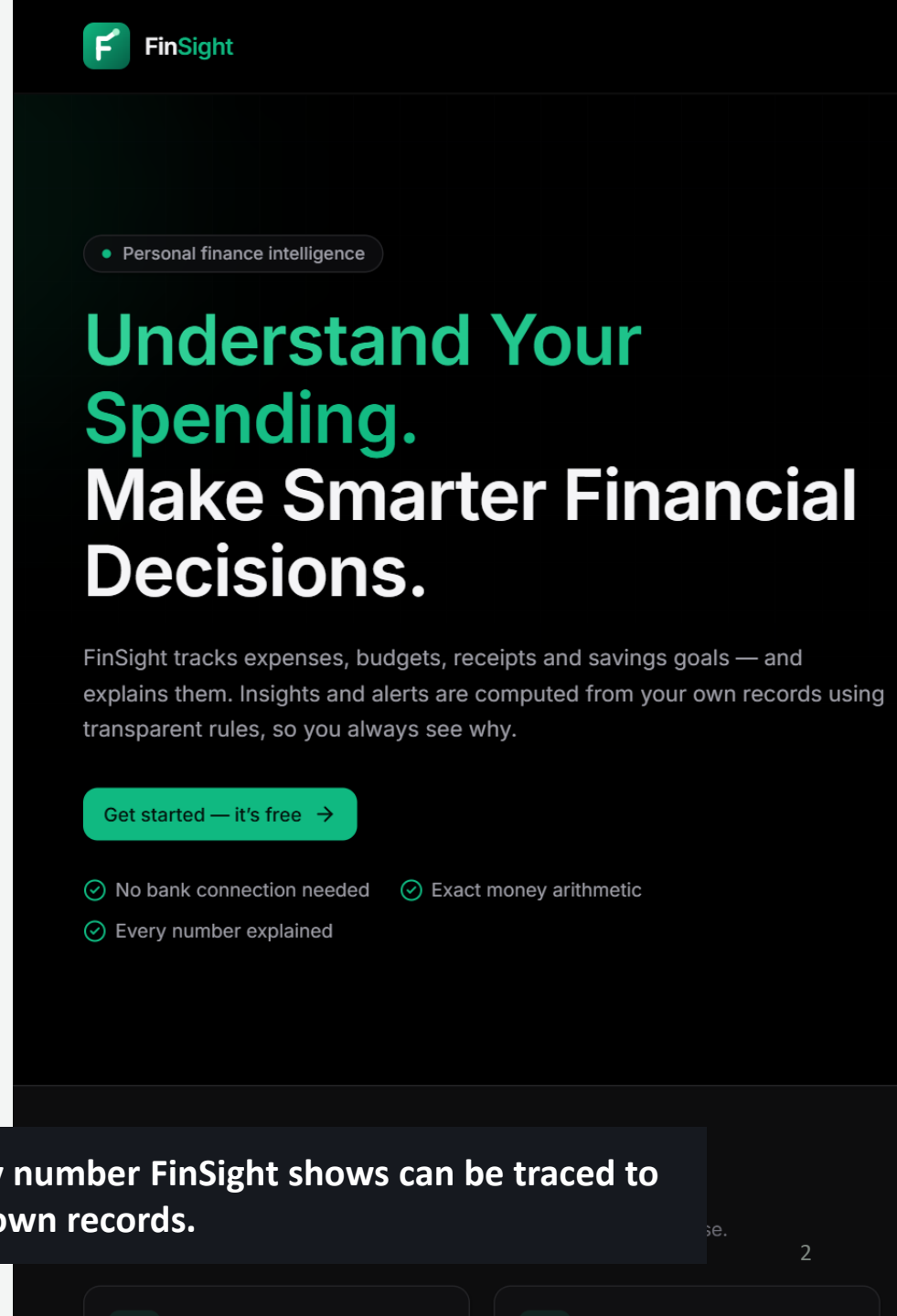
Paper receipts must be typed in by hand, so many purchases are never recorded.

Unexplained numbers

Charts show amounts, not the calculation or transactions behind them.

Late discovery

Overspending and duplicate charges are noticed only at month end.



Goal: every number FinSight shows can be traced to the user's own records.

OBJECTIVES AND KEY FEATURES

ELEVEN OBJECTIVES · ALL IMPLEMENTED AND VERIFIED

Expenses

Add, edit, search, filter, sort and export, in exact paise

Budgets

Overall and category budgets, one alert per threshold

Analytics

Trends, categories, payment methods, merchants

Insights

Labelled fact, estimate or projection, with evidence

Anomalies

Robust statistics flag charges for review only

Receipt OCR

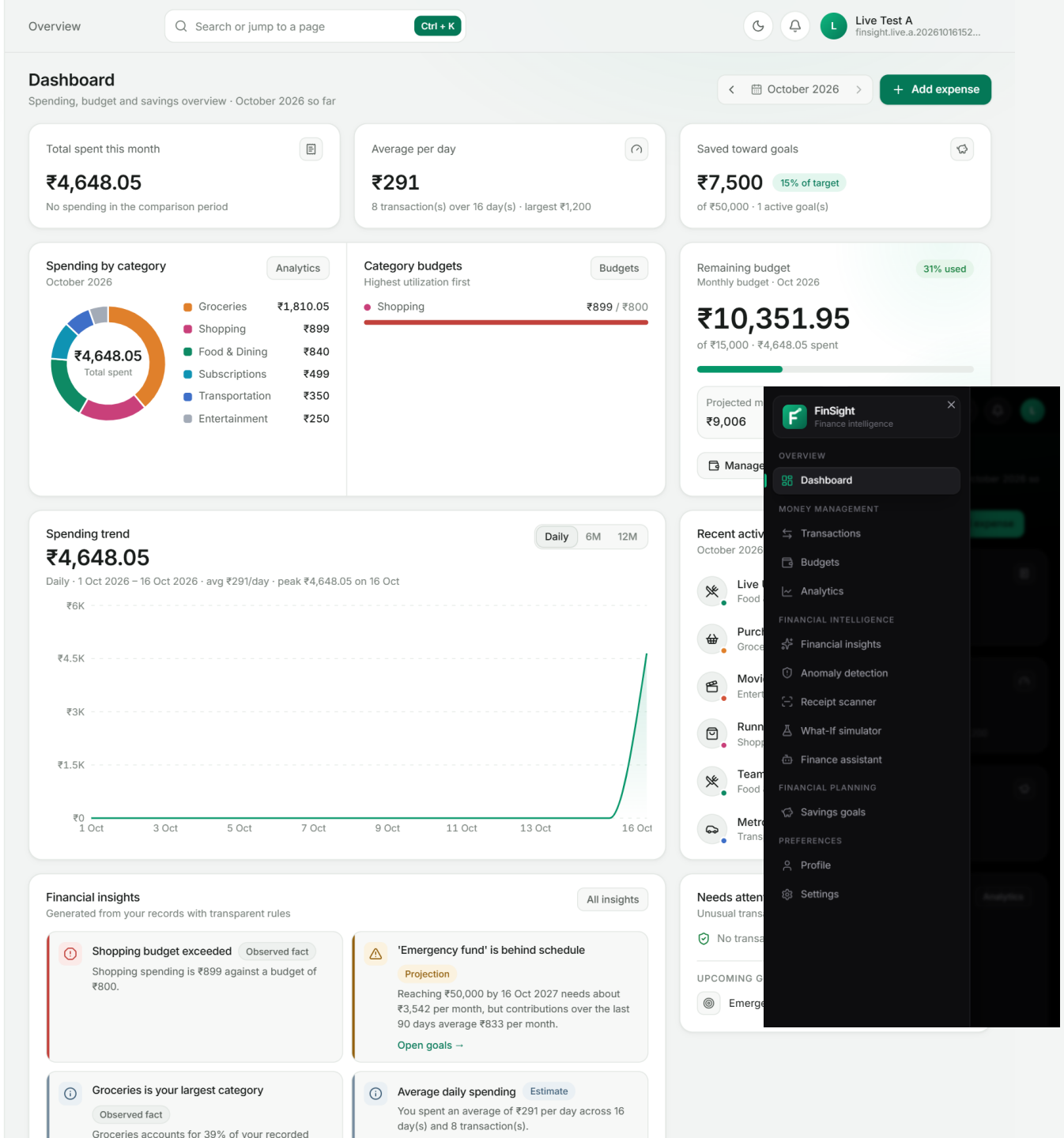
Tesseract reads the receipt; the user confirms

What-If

Up to four saving scenarios, no side effects

Goals & assistant

Savings goals plus a rule-based assistant



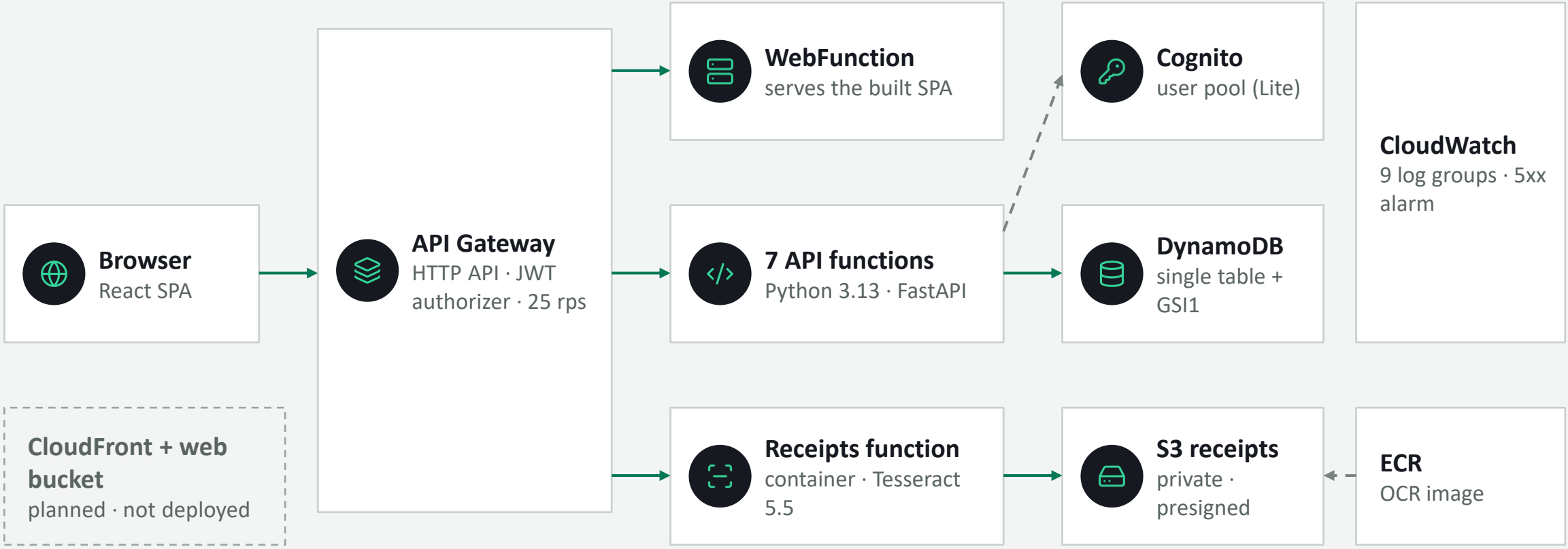
APPLICATION OVERVIEW

LIVE SCREENS · SYNTHETIC TEST USER

- 16 routes: dashboard, transactions, budgets, analytics, insights, anomalies, receipts, What-If, assistant, goals
- Cognito sign-in; every record scoped to the signed-in user in DynamoDB
- Dark (default) and light themes; mobile drawer below 1024 px
- Animations that respect reduced motion
- “Created by Aakar Gupta” credit and an About-the-author dialog

ACTUAL SYSTEM ARCHITECTURE

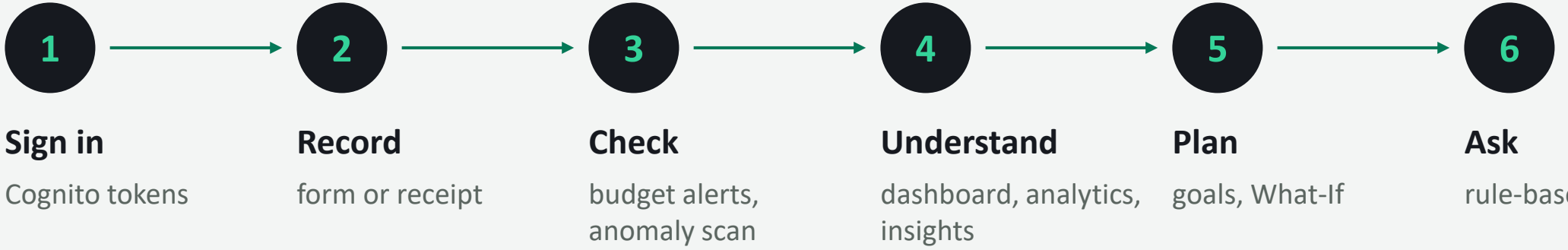
AS DEPLOYED · STACK FINANCE-INTELLIGENCE-PROD · AP-SOUTH-1

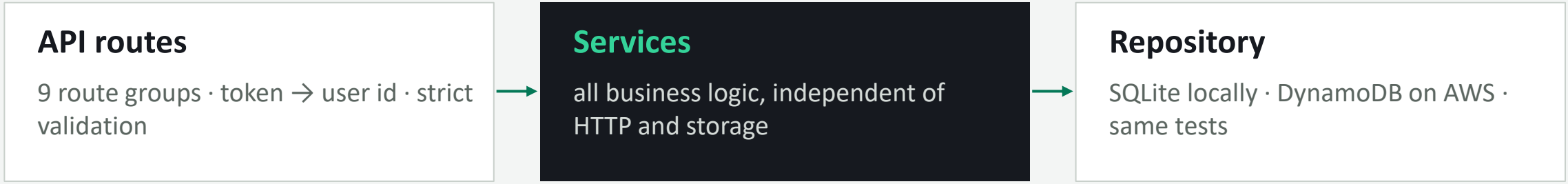


One origin for the SPA and /api/v1 (no CORS) · browser uploads receipts straight to S3 with presigned URLs · one IAM role per function · no VPC, NAT, EC2 or RDS

CORE MODULES AND WORKFLOW

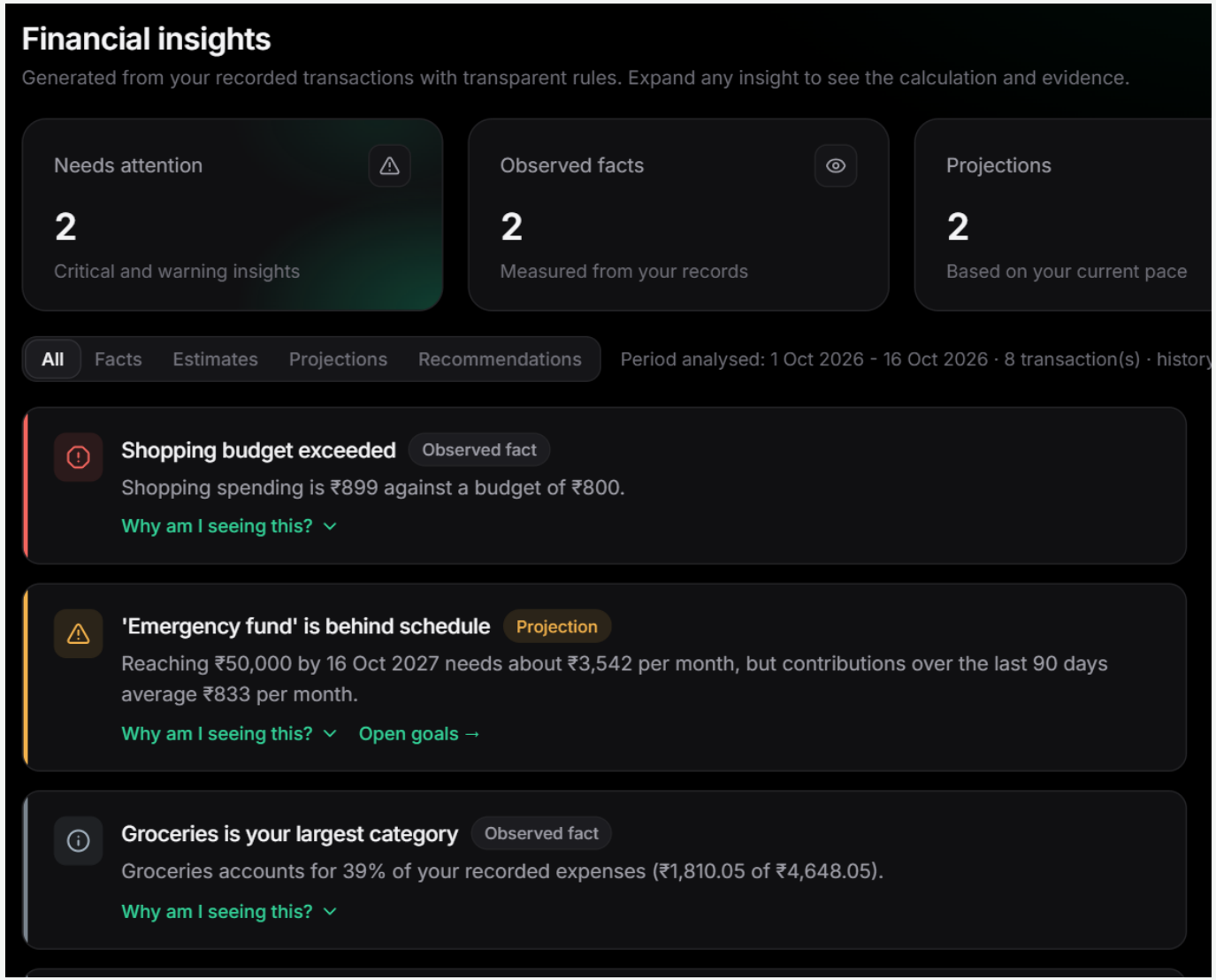
ONE USER JOURNEY · ONE LAYERED CODEBASE





Modules Expenses · Budgets · Analytics · Insights · Anomalies · Receipts · Simulator · Goals · Assistant

Each Lambda builds only its own route group, so it receives only the IAM permissions that group needs.



ANALYTICS AND INTELLIGENCE

EVERY NUMBER IS EXPLAINED

+ INSIGHT ENGINE

Each insight is labelled fact, estimate, projection, recommendation or data gap, with its period, formula, values and transactions.

+ ANOMALY DETECTION

Modified z-score $0.6745 \cdot (x - \text{median}) / \text{MAD} \geq 3.5$ and amount $\geq 2.5 \times$ the category median. Flags are reviewed; data is never changed.

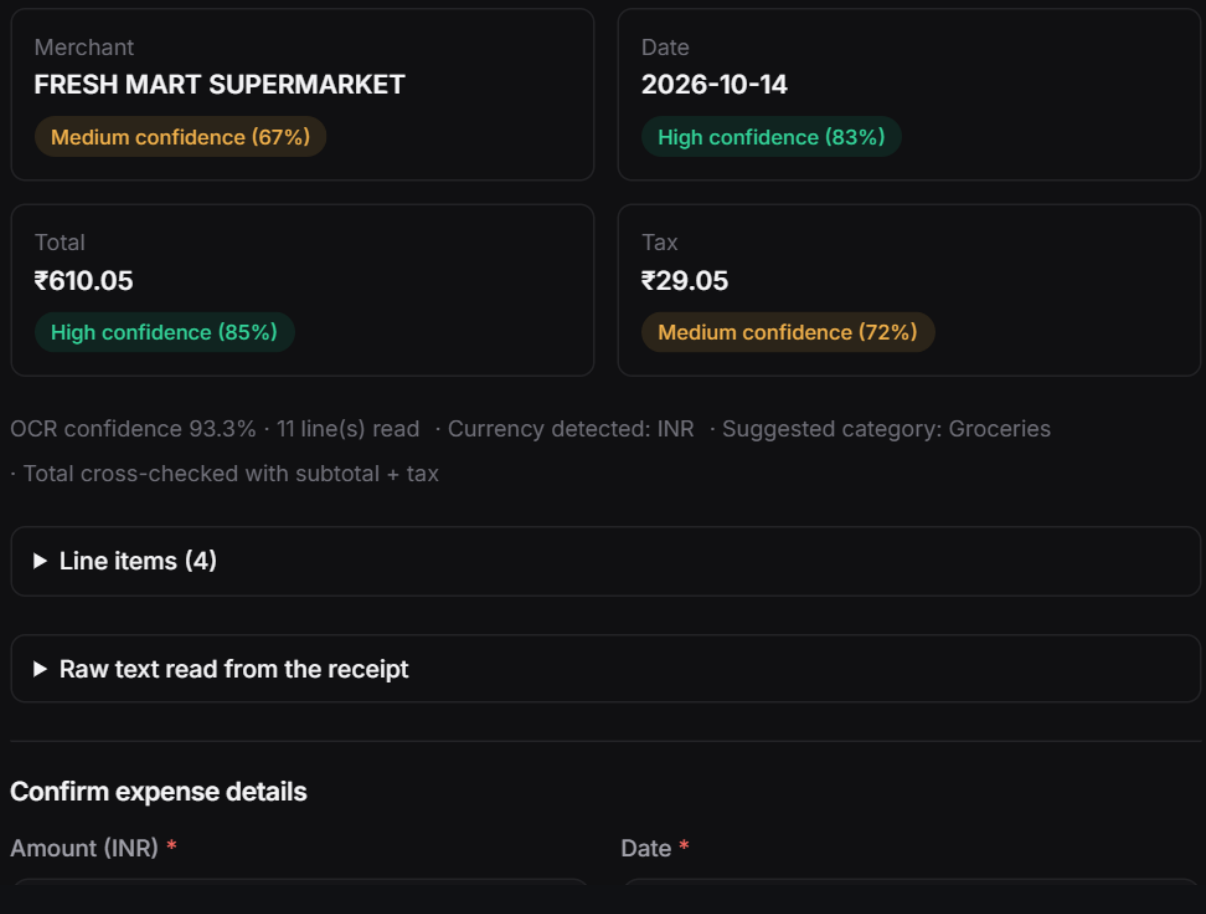
+ BUDGET PROJECTION

Month-end spend = spent ÷ days elapsed × days in month. Each threshold (75 / 90 / 100 %) alerts once per month.

RECEIPT OCR WORKFLOW

PRESIGNED UPLOAD · OCR IN LAMBDA · REVIEW

1. **Request upload URL**
   5-minute presigned POST fixes key, type, encryption and 1 B–5 MB size
2. **Upload to private S3**
   the browser posts the image straight to the bucket
3. **OCR in Lambda**
   validation, Pillow preprocessing, Tesseract --psm 4
4. **Parse with confidence**
   merchant, date, total, tax, currency, line items
5. **Review and confirm**
   the user corrects fields; a linked expense is created



93.3 % OCR confidence

₹610.05 total extracted

3.2 s processed in Lambda

AWS DEPLOYMENT AND SECURITY

LIVE SINCE 16 OCTOBER 2026 · AWS SAM

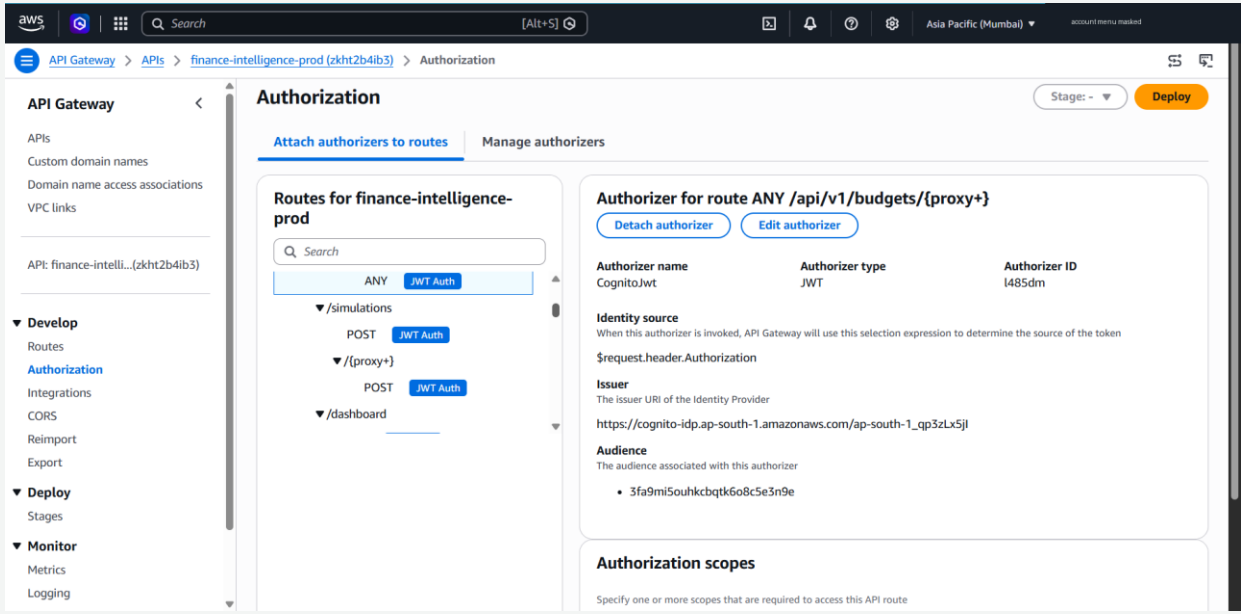
STACK	**finance-intelligence-prod · 63 resources**
HOSTING	**API Gateway + WebFunction (CloudFront blocked)**
RELEASES	**change-set review · 3 code-only updates**

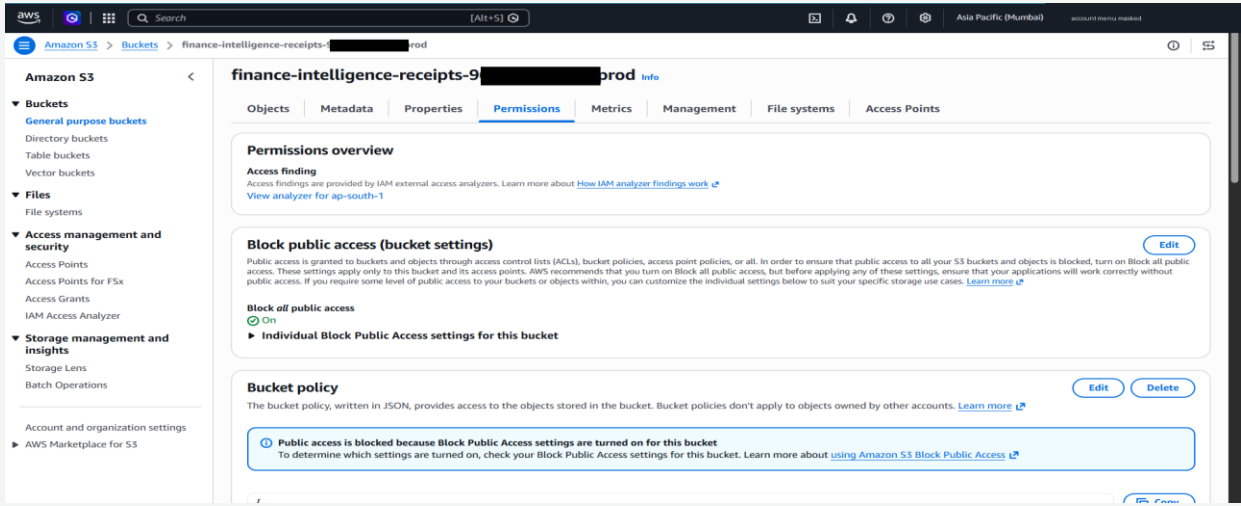
Two token checks JWT authorizer, then JWKS and Cognito GetUser (sign-out revokes within 15 s)

Per-user isolation user id only from the token; other users' ids return 404

Private receipts Block Public Access, SSE-S3, TLS-only, presigned URLs

Least privilege one IAM role per function; no Scan, no wildcards





AWS Management Console screenshots · account ID masked

TESTING, EVALUATION AND COST

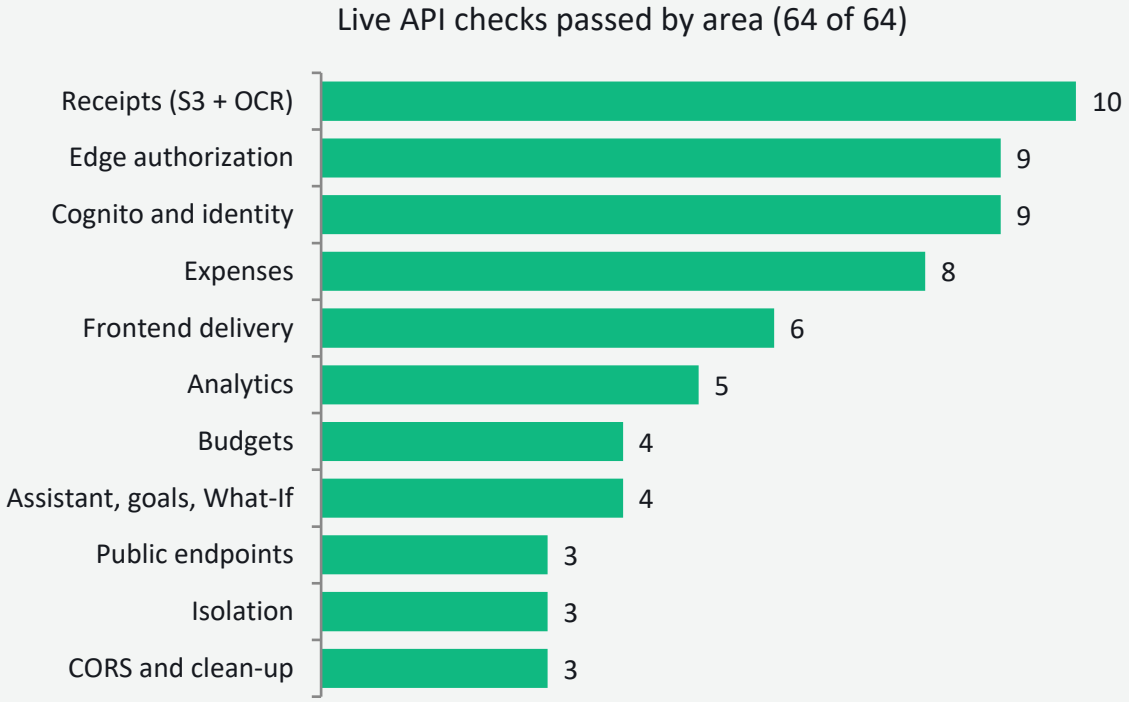
FINAL RUNS · 16 OCTOBER 2026

64/64
live API, Cognito, S3, OCR checks

27/27
live browser checks · 0 errors

131
backend tests passed

20
frontend tests passed



ESTIMATED COST

≈ US$0.05–0.10 / month

- Free plan, US$159.91 credits; US$0.09 used
- Largest idle cost: ECR images (~0.38 GB)
- No EC2, RDS, NAT or load balancer
- An estimate for demo use, not a guarantee

CONCLUSION AND FUTURE WORK

WHAT WAS DELIVERED · WHAT REMAINS

+ CONTRIBUTIONS

- Full finance platform: expenses, budgets, analytics, insights, anomalies, OCR, What-If, goals, assistant
- Explainable by design: every insight shows its evidence
- Serverless on AWS with two-layer auth, revocation and per-user isolation

+ LIMITATIONS

- CloudFront blocked until AWS verifies the account
- Revocation within 15 s; fail-open if Cognito is down
- Cold starts ~3 s; no load test or OCR accuracy study

+ FUTURE WORK

- CloudFront + WAF, then a Content-Security-Policy
- IAM Identity Center for deployments; SNS alarms; DynamoDB PITR
- Statement import and better OCR preprocessing

LIVE https://zkht2b4ib3.execute-api.ap-south-1.amazonaws.com